AUTODESK
Instructables

Quiz the Spire

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Introduction: Quiz the Spire

Supplies

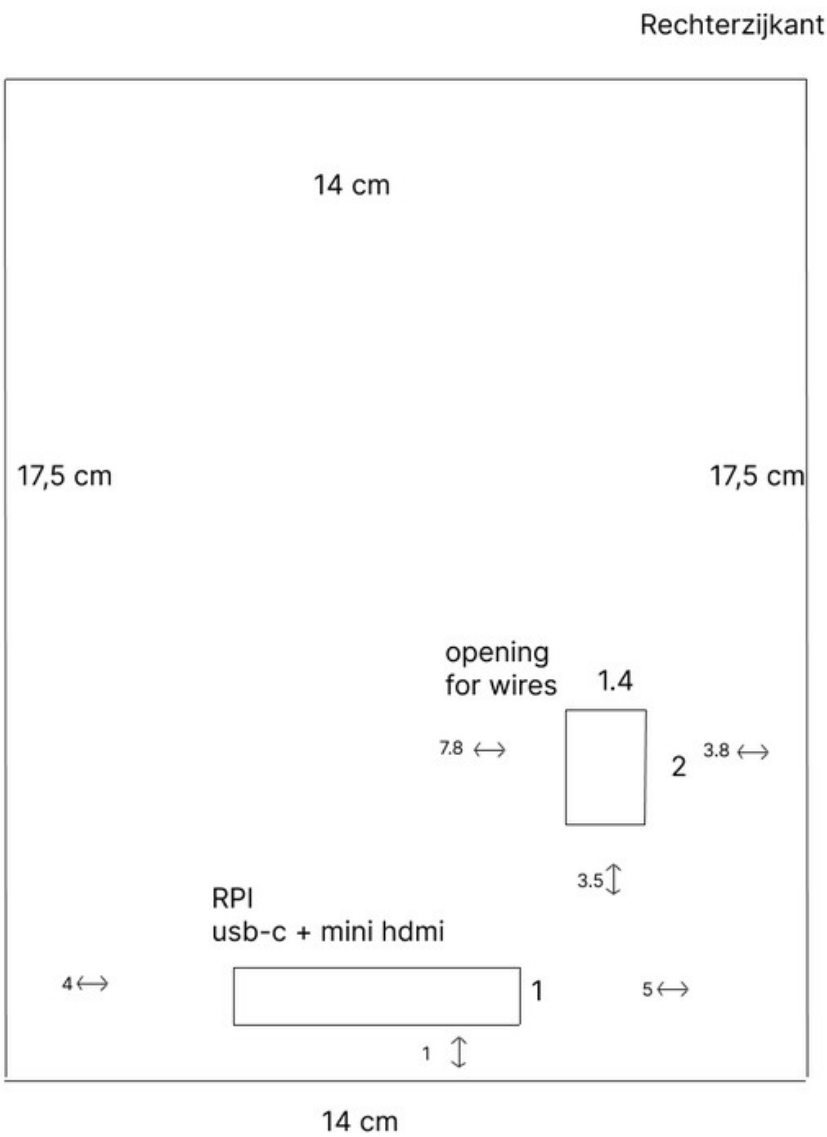
Materials Needed

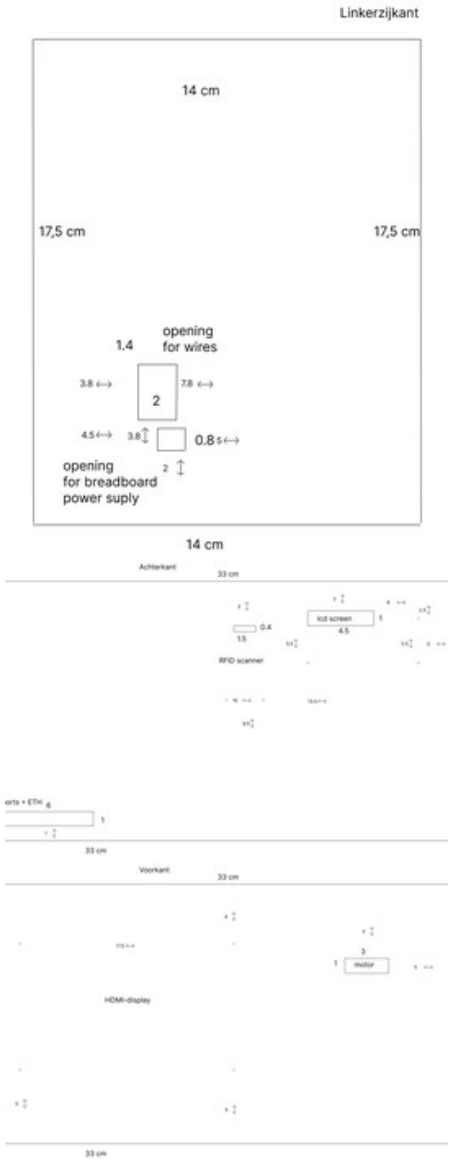
1. Sandpaper
2. Wood drill
3. Screws
4. Screwdriver
5. Metal backplate
6. Small HDMI screen
7. With screw holes
8. USB-C power support
9. silicone adhesive
10. Wooden box (to house the components)
11. **Raspberry Pi 5**
12. **16x2 LCD**
13. **Servo motor**
14. **2 breadboards**
15. **Breadboard power supply**
16. **18B20 one-wire temperature sensor**
17. **Photoresistor** (light-diminishing resistor)
18. **SNES-style controller (Raspberry Pi compatible)**
19. **Wood rasp**

Step 1: Measure First

1. Use a ruler or calliper to measure the exact dimensions of each component.
2. Don’t forget to factor in space for wires, especially GPIO connectors and USB cables.

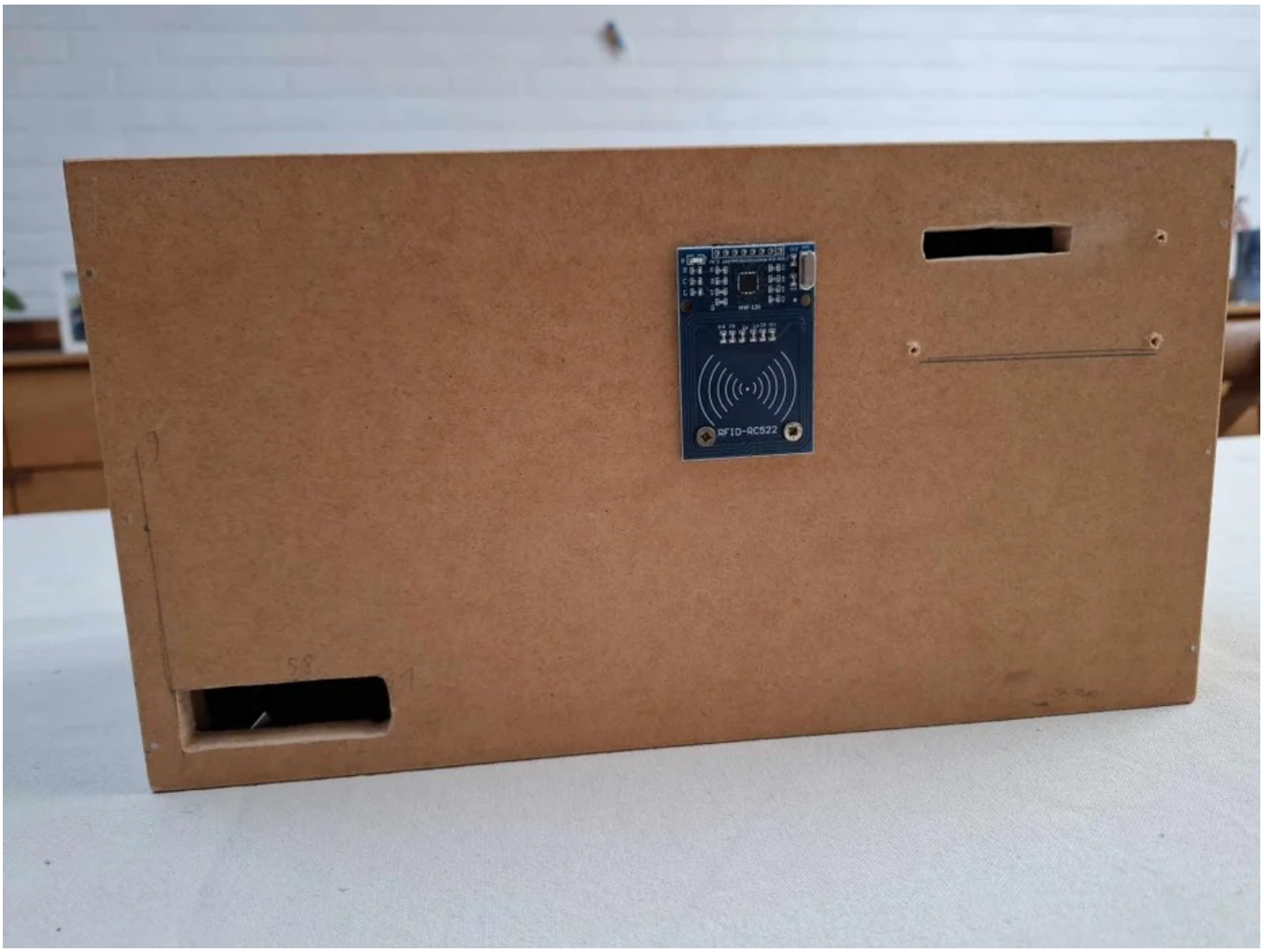
Step 2: Mark the Spots





1. Use a pencil to draw the locations of each hole on the box lightly.
2. Ensure logical placement: e.g. screen on front, GPIO access, screws.

Step 3: Drill Starter Holes



1. Use a wood drill to create the base holes at the marked positions.

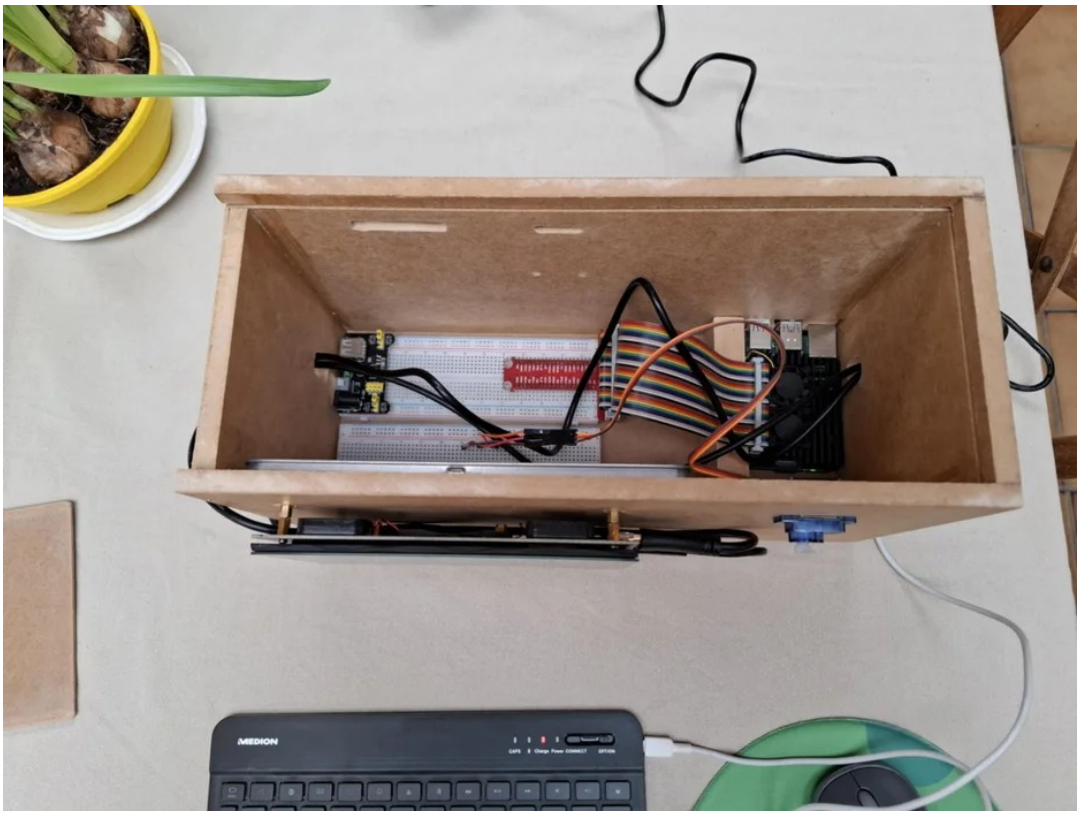
Step 4: Shape With Wood Rasp

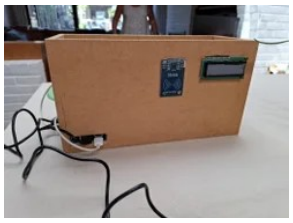
1. Use the wood rasp to carefully enlarge and shape the holes to match your components.
2. Aim for a snug fit, but leave enough room for wiring, especially for GPIO pins and connectors.

Step 5: Attach a Metal Plate With Epoxy

1. Attach the metal backplate to the back of the front pannel of the box of where you're planning on putting your HDMI screen using silicone adhesive.

Step 6: Test Fit Components





1. Insert each component temporarily to ensure fit.
2. Make adjustments as needed before moving on.

Step 7: Put Code on Raspberry Pi 5

1. Put the code on the raspberry pi 5 from this github:

Step 8: Run App.py or Make It Start by Default

To run the backend you must run the app.py file using the virtual environment.

Put the frontend as your apache2 root directory.

Optionally but recommended create a system service on your raspberry pi that automatically starts appmonitor_backup.py as user root, this will start up the browser to the apache2 site in fullscreen, map the shoulder buttons to shutting down the pi, run app.py using the venv.

You will need to change the directories in the files.

Step 9: Install Ydotool and Set Up Gamepad Script Autostart (optional)

Goal: Install input automation tool ydotool, set up your Python environment, and auto-launch gamepad.py on boot.

1. ✅ Install ydotool

sudo apt update

sudo apt install cmake g++ libevdev-dev libudev-dev libyaml-cpp-dev git -y

git clone https://github.com/ReimuNotMoe/ydotool.git

cd ydotool

cmake .

make

sudo make install

(ydotool is tricky to use, if you have any errors, ask Deepseek)

🔧 *Note:* ydotoold must be run as root or with appropriate permissions, so keep that in mind for autostart.

2. 🐍 Set Up Python Virtual Environment

sudo apt install python3-venv -y

mkdir ~/gamepad_project

cd ~/gamepad_project

python3 -m venv venv

source venv/bin/activate

pip install -r requirements.txt

3. 📜 Gamepad.py Script

1. Make sure it's executable:

chmod +x gamepad.py

4. 🔁 Auto-Start gamepad.py on Reboot Using tmux

Step 1: Install Tmux

sudo apt update && sudo apt install tmux -y

Step 2: Add Auto-Start Script

Edit your user's .profile file:

nano ~/.profile

Paste this at the bottom (replace paths with yours):

```
# Auto-start gamepad.py in tmux

if [ -n "$PS1" ] && ! tmux has-session -t gamepad 2>/dev/null; then

tmux new -d -s gamepad -c ~/Project/project-one/backend/raspberryPi5

tmux send-keys -t gamepad "source ~/Project/.venv/bin/activate && python gamepad.py" Enter

fi
```

Save: Ctrl+O → Enter → Ctrl+X

Reload: source ~/.profile

Step 3: Test It

1. **Log out and log back in** (or reboot).
2. **Check if it's running**: tmux list-sessions

Should show: gamepad: 1 windows (created ...)

View output (optional): tmux attach -t gamepad

Troubleshooting

1. **Not starting?** Manually test the script first:

```
cd ~/Project/project-one/backend/raspberryPi5

~/Project/.venv/bin/python gamepad.py
```

Permission issues? Ensure your user owns the files:

```
sudo chown -R $USER:$USER ~/Project
```